

Amateur Astronomy

The Bangalore Astronomical Society®



Overview

- What is Amateur Astronomy?
 - Amateur Astronomy as a hobby
 - We wish to share the joys of Amateur Astronomy with you
 - Scientific Contributions by hobby astronomers
 - Various means of contributions
 - Why these are important
 - What you can do
 - Some ideas for students
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Amateur Astronomy - What?

- What is Amateur Astronomy?
 - Amateur, Not Professional, a Hobby
 - May have productive outcome
 - Observing celestial objects
 - Photography of the sky and celestial objects
 - Observing “variable” stars
 - Hunting down new comets, asteroids and variable stars
 - Observing meteor showers, aurorae, and other such atmospheric phenomena
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What professionals do

- Explain physically, the origin and characteristics of celestial objects and events
 - How do galaxies form? Make a physical model...
 - Why does this star change in brightness randomly?
 - Why are the planets' orbits confined more-or-less to a single plane?
 - What are comets made of?
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What professionals do

- Cosmology
 - Trace back to the origins of the universe
 - Scientific history of the universe
 - Dark Matter and Dark Energy
 - Cosmic Microwave Background Radiation
 - Data Collection
 - Collect data for future analysis and physical modeling
 - Eg: Variation in brightness of stars
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What amateurs do

- Have fun – a hobby
 - See beautiful nebulae, galaxies etc and enjoy
- *Learn*, at the same time, by doing
 - How telescopes work, optics of telescopes
 - Telescope making – make your own telescope from scratch
 - Observe features of nebulae, galaxies, star clusters etc and learn why they are that way

What amateurs do

- Contribute to scientific research
 - Find comets and asteroids, so professionals can study them
 - Hunt for supernovae, so that professionals can find what's going on there
 - Collect data on variation of brightness of stars
 - Recover “lost” comets
 - Lots more...
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Amateur Astronomy – a Hobby

- Observation – look at the sky, the celestial objects and atmospheric phenomena
 - Amateur Astrophotography – take photographs, of the sky, and the celestial objects
 - Amateur Telescope making – make telescopes and other instrumentation
 - Reducing light pollution?
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What you get to see

- *Not as dramatic as the photos though, unfortunately!*





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Astrophotography

- What you need
 - A camera capable of long exposures
 - A tripod
- Scientific Value
 - Comet, asteroid and nova discovery
 - Variable Stars



Constellation Scorpius by Shashank H.J.

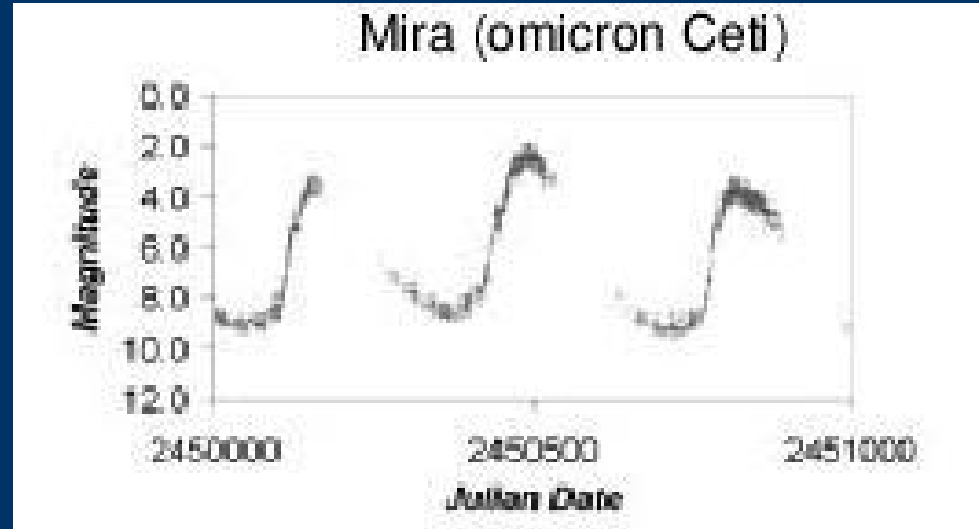


Eagle Nebula by Dr. Suresh Mohan

Scientific Work from Amateurs



Comet Hale Bopp, discovered by Thomas Bopp and Alan Hale (also a professional).



Light curve of Mira, a variable star frequently observed by amateurs



What YOU can do!

- Try observing the motion of planets across the sky (Retrograde and Prograde motion)
 - How does the time of moonrise change everyday
 - Find the orbital period of the moon
 - Calculate the distance to the moon! (How?)
 - When does the sun rise on the summer solstice and winter solstice?
 - Find your latitude! (How?)
 - Look up on the cosine rule for a triangle on a sphere!
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What YOU can do!

- Variable Stars: Find the distance to stars!!!
 - Observe and find the time periods of Cepheid Variables
- Plot light curves of various variable stars
 - <http://www.aavso.org> - Online resource

What YOU can do!

(From dark skies)

- Look around for fuzzy patches in the sky
 - What are these fuzzy patches?
 - Observe the Milky Way
 - Why do we see the Milky Way, our galaxy, like a band in the sky?
 - Estimate your Naked Eye Limiting Magnitude and the Bortle Sky Class (google these terms down!)
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What YOU can do!

(With a telescope)

- Find the mass of Jupiter!
 - How?
 - Find the speed of light!
 - Observe eclipses of Jupiter's moons and predict an eclipse six months later.
 - Observe the actual eclipse six months later
 - How????
 - What are those black spots in the nebula?
 - Jeans Instability Condition
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What YOU can do!

(With a camera)

- Take photographs of the sky
 - What is the faintest star in the photograph?
 - Is there some way of finding out the magnitudes of stars photographically?
- Learn how photography works – camera optics
 - How much of the sky can my camera cover at a time?
 - How would you calculate this?

If you are really interested...

- Try to get your school's telescope ready and use it!
- Join us:
 - <http://www.bas.org.in>

Thank You
